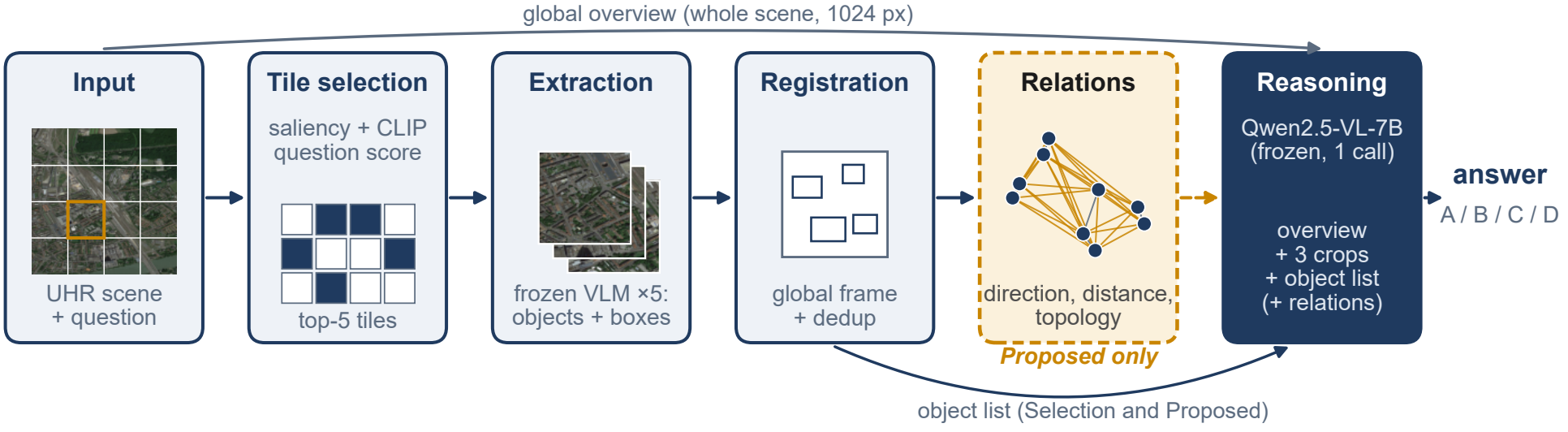

global overview (whole scene, 1024 px)
Input
UHR scene
+ question
Tile selection
saliency + CLIP
question score
top-5 tiles
Extraction
frozen VLM ×5:
objects + boxes
Registration
global frame
+ dedup
Relations
direction, distance,
topology
Proposed only
Reasoning
Qwen2.5-VL-7B
(frozen, 1 call)
overview
+ 3 crops
+ object list
(+ relations)
answer
A / B / C / D
object list (Selection and Proposed)